An Exact First-Passage Atlas for Twenty Frozen Subgroup Targets

Anonymous

Abstract

We determine the complete first-passage law for each of the twenty actual subgroups in a frozen sixteen-label closure model. A uniformly random label ordering produces prefix supports A_k, and the target time is $T_H = \min\{k : H \le \Phi(A_k)\}$. Exact subset counts give the cumulative law, while pivotal prefix edges give every ordered-permutation mass by a second formula. The trivial target is hit at time zero; the top target recovers the previously frozen full-assembly law entry by entry and has mean 36499/3960. Subgroup inclusion orders all twenty variables pointwise and hence in stochastic order; all 102 comparable ordered pairs are certified. Independent minimal-antichain reconstruction, symbolic rational checks, clean replay, and forty hostile mutations audit the result. The scope is finite and obeys `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Frozen model and hitting event

Let $L = \{S_1, \dots, S_{16}\}$ be the source-bound labels and let

$$\Phi(A) \le Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$$

be the subgroup generated by $A \subseteq L$. The frozen C75 order lists the twenty actual subgroups as $H_0, \dots, H_{19}$, with $H_0 = \{0\}$ and $H_{19} = Q$. If π is uniform among the 16! label permutations, let A_k be its first k labels, including $A_0 = \varnothing$, and define

$$T_H(\pi) = \min\{k : H \le \Phi(A_k)\}.$$

This is first passage into the containment up-set above H. It is not the event $\Phi(A_k) = H$; exact-closure hits and skips are not part of this atlas.

For $0 \le k \le 16$, put

$$F_H(k) = \#\{A \subseteq L : |A| = k,\ H \le \Phi(A)\}.$$

For a hitting support S, let

$$p_H(S) = \#\{\ell \in S : H \not\le \Phi(S \setminus \{\ell\})\}$$

be its number of pivotal labels.

2 Exact law and stochastic order

Theorem 1 (Subgroup first-passage atlas)**.** *For every actual target H and $0 \le k \le 16$,*

$$\Pr(T_H \le k) = \frac{F_H(k)}{\binom{16}{k}}, \tag{1}$$

$$\Pr(T_H > k) = 1 - \frac{F_H(k)}{\binom{16}{k}}. \tag{2}$$

For $k \ge 1$, the exact number $N_H(k)$ of permutations first hitting at time k is

$$N_H(k) = \sum_{\substack{|S|=k \\ H \le \Phi(S)}} p_H(S)(k-1)!(16-k)!. \tag{3}$$

Moreover,

$$\mathbb{E}[T_H] = \sum_{k=0}^{15} \left(1 - \frac{F_H(k)}{\binom{16}{k}}\right). \tag{4}$$

If $H \le K$, then $T_H(\pi) \le T_K(\pi)$ for every permutation π. Thus their CDFs, survival functions, and expectations obey the corresponding stochastic order.

Proof. The unordered size-k prefix of a uniform permutation is uniform among the $\binom{16}{k}$ subsets, proving (1); complementing gives (2). A permutation first hits at $k \geq 1$ exactly when its prefix support S hits and its last prefix label is pivotal. For each such pair (S, ℓ), the preceding members have $(k-1)!$ orders and the remaining labels have $(16-k)!$ orders, proving (3). Summing the integer tail gives (4). Finally, any prefix closure containing K also contains $H \leq K$, so the pointwise order follows. ☐

For H_0, all 16! permutations have $T_{H_0} = 0$. For $H_{19} = Q$, the complete row of subset counts, pivotal data, first-passage masses, reduced probabilities, survival counts, and expectation agrees entrywise with the final frozen full-assembly certificate.

3 Exact finite inventory

Table 1 records each target order, attainable time range, minimal hitting-support count $|\mathcal{M}_H|$, and exact expectation. The canonical receipt additionally stores every one of the 20×17 exact count, probability, and survival cells, all pivotal patterns, each minimal support mask, and a complete 65536-bit hit indicator for every target.

i	$\lvert H_i\rvert$	range	$\lvert\mathcal{M}_i\rvert$	$\mathbb{E}T_i$	i	$\lvert H_i\rvert$	range	$\lvert\mathcal{M}_i\rvert$	$\mathbb{E}T_i$
0	1	0–0	1	0	10	9	2–10	38	13243/3960
1	2	1–16	1	17/2	11	9	1–13	18	289/72
2	3	1–10	24	3961/1320	12	9	1–13	18	289/72
3	3	1–9	13	2363/990	13	9	1–13	13	34/9
4	3	1–10	24	3961/1320	14	18	3–16	38	2363/264
5	3	1–10	30	12631/3960	15	18	2–16	18	12121/1320
6	6	2–16	24	35207/3960	16	18	2–16	18	12121/1320
7	6	2–16	13	8687/990	17	18	2–16	13	4522/495
8	6	2–16	24	35207/3960	18	27	2–13	25	1513/360
9	6	2–16	30	357/40	19	54	3–16	25	36499/3960

Table 1: Summary of the twenty exact first-passage rows.

The C85 point-set inclusion matrix has 102 ones, including the twenty reflexive pairs. For every such pair (i, j), the certificate checks the stronger subset implication

$$H_j \leq \Phi(A) \implies H_i \leq \Phi(A) \qquad (H_i \leq H_j)$$

for all 65536 supports. The CDF, survival, and expectation comparisons are then checked explicitly as redundant exact consequences.

4 Independent reconstruction and audit

The producer builds an indexed transition table from the 54-point addition law and enumerates all support closures. It obtains each $N_H(k)$ both from successive exact CDF values and from (3). The checker takes a separate representation-level route: it expands finite point sets directly, discovers every inclusion-minimal support in

$$\mathcal{M}_H = \min_{\subseteq}\{S : H \leq \Phi(S)\},$$

and reconstructs a hit solely from the condition that it contains some $M \in \mathcal{M}_H$. The complete canonical objects agree.

A SymPy route verifies twenty normalized probability generating functions, their exact derivative means, all pivotal-polynomial coefficients, and the 102 stochastic-order pairs. Clean-process replay preserves the evidence SHA-256

`4511d434f477784782f2af5106afff4c2cf3b48cd7eb7a62ed05b8f2f42afb1b`.

All 40 hostile semantic mutations are rejected.

This finite theorem makes no claim about arithmetic or local data, Euler factors, root numbers, automorphy, a full Burnside ring or table of marks, or a Hilbert–Polya operator.